Supplementary Figures

EMToolbox uncovers biased co-option of embryonic EMT programs across cancers

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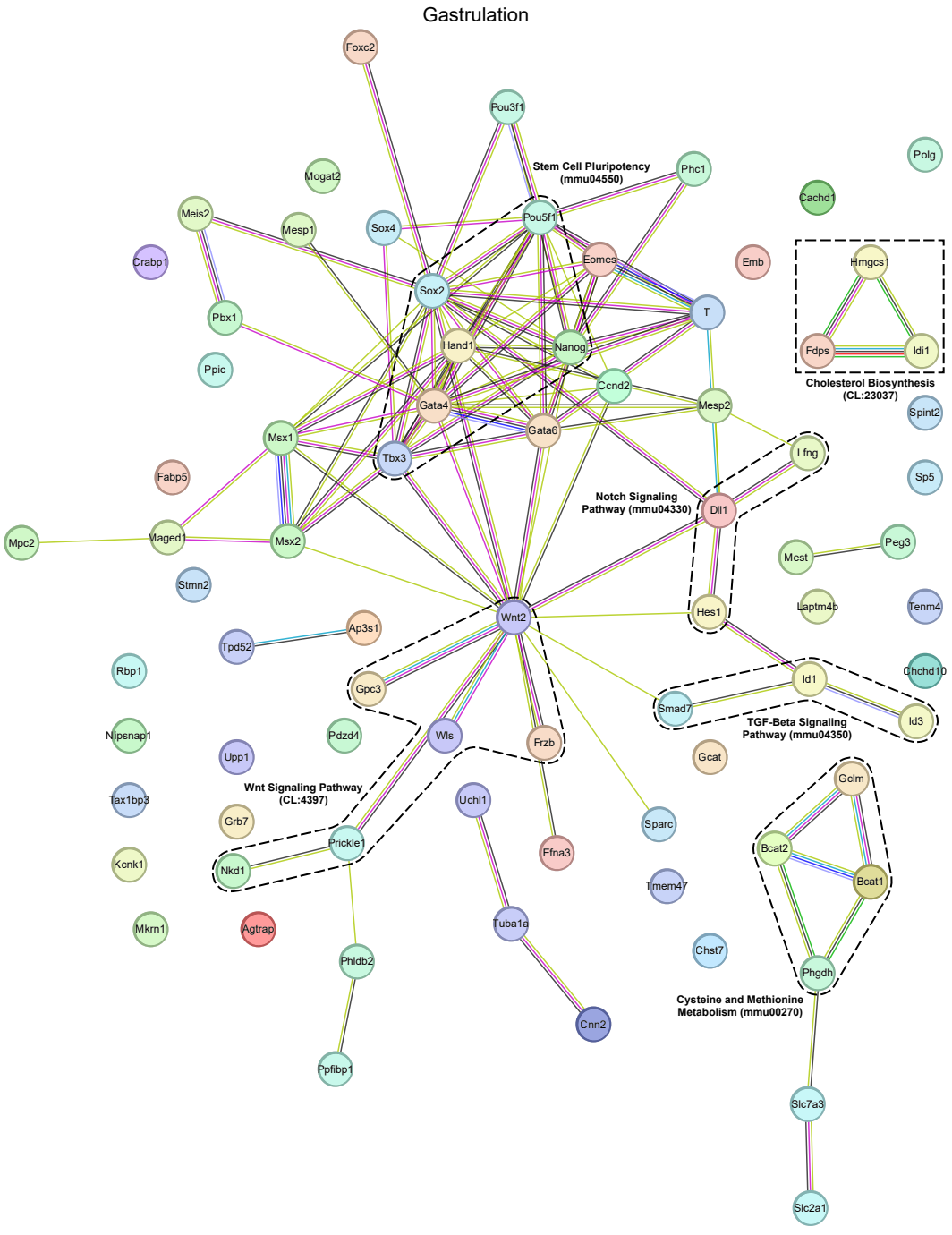
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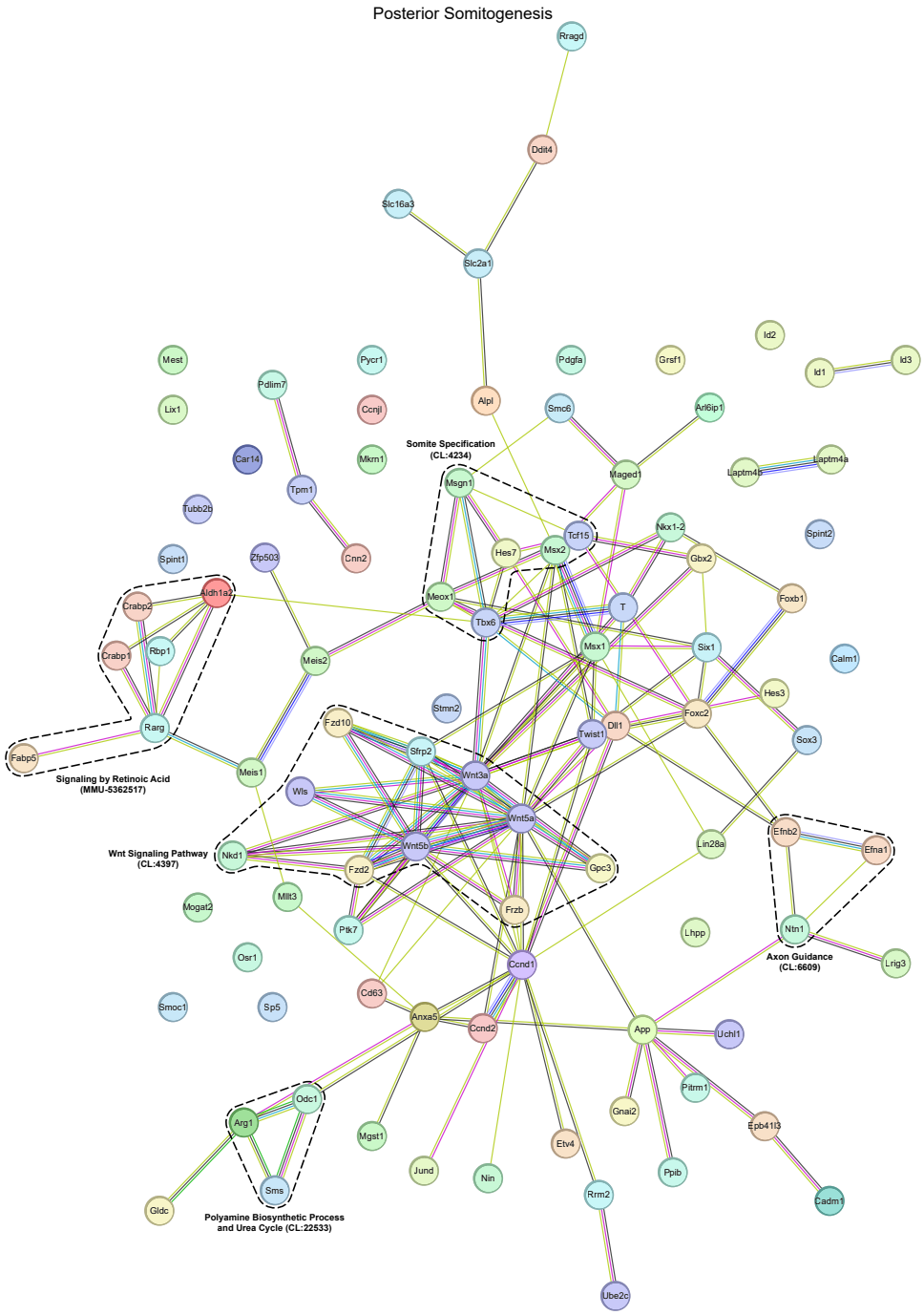
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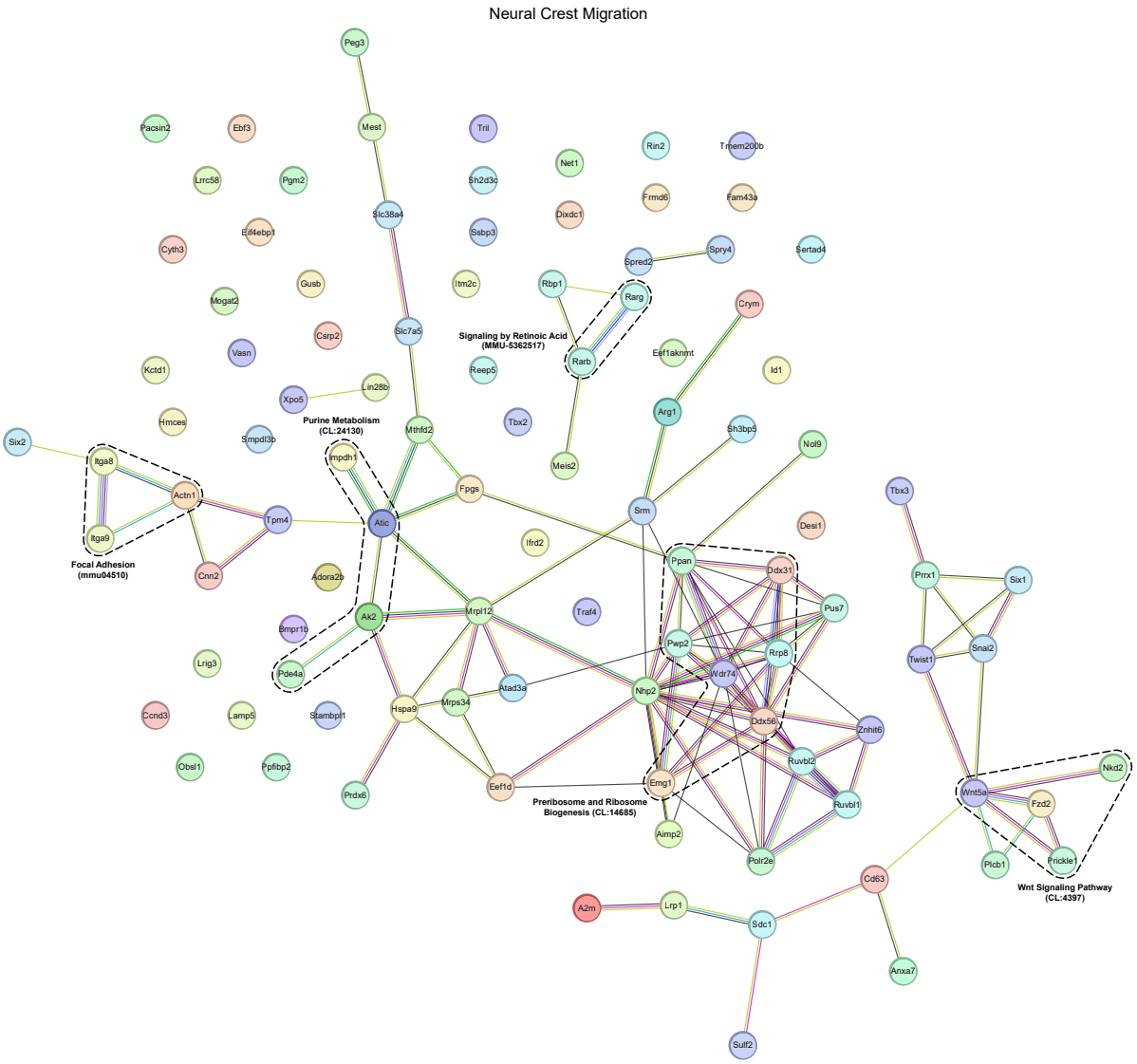
Supplementary Figure 1: STRING network for Gast

STRING protein-protein interaction network for Gast. Dashed outlines indicate functional enrichments, including Wnt, Notch, and TGF-β signaling.



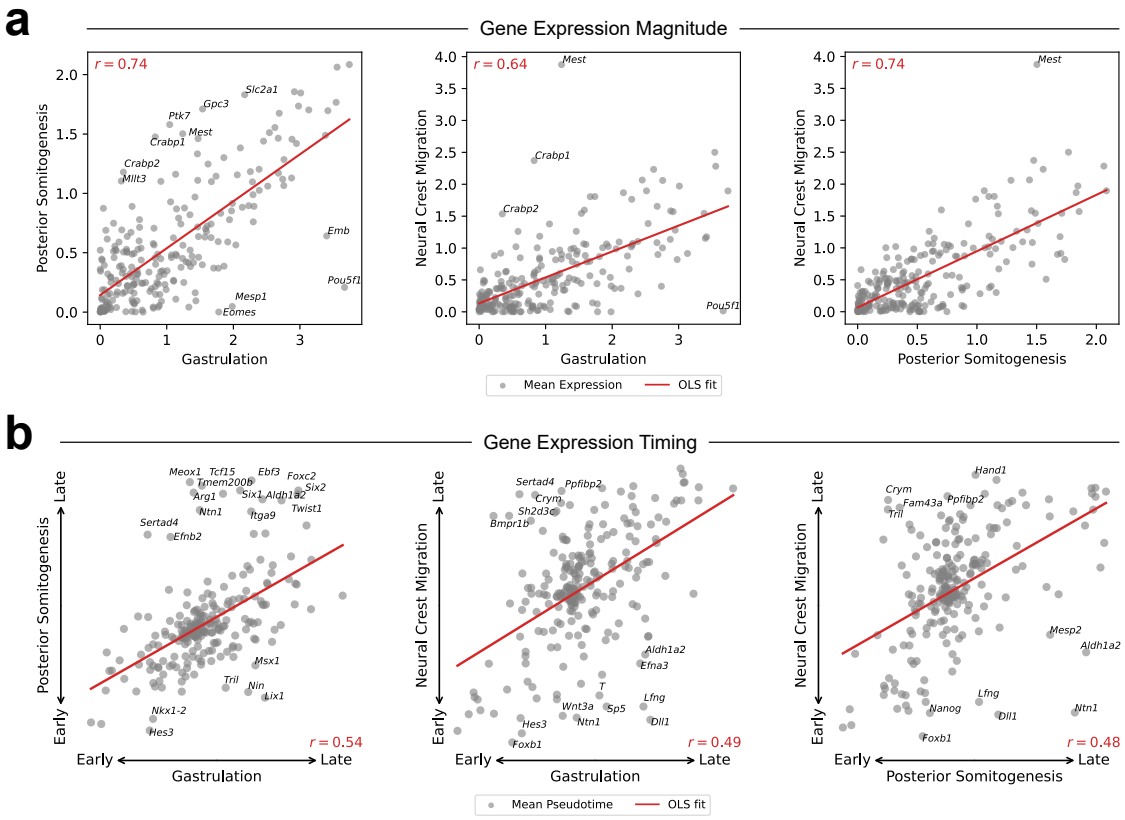
Supplementary Figure 2: STRING network for pSom

STRING protein-protein interaction network for pSom. Dashed outlines indicate functional enrichments, including Wnt and retinoic acid signaling.



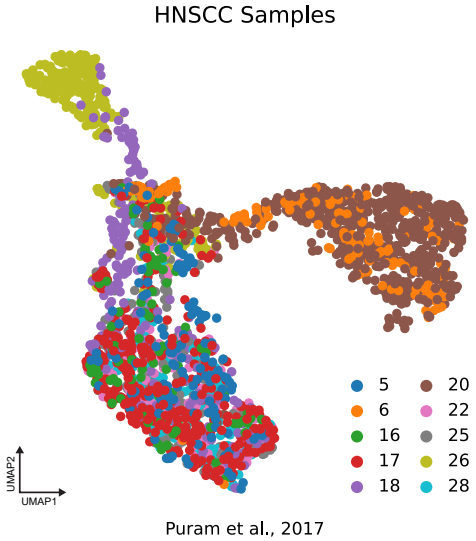
Supplementary Figure 3: STRING network for Crest

STRING protein-protein interaction network for Crest. Dashed outlines indicate functional enrichments, including Wnt and retinoic acid signaling.



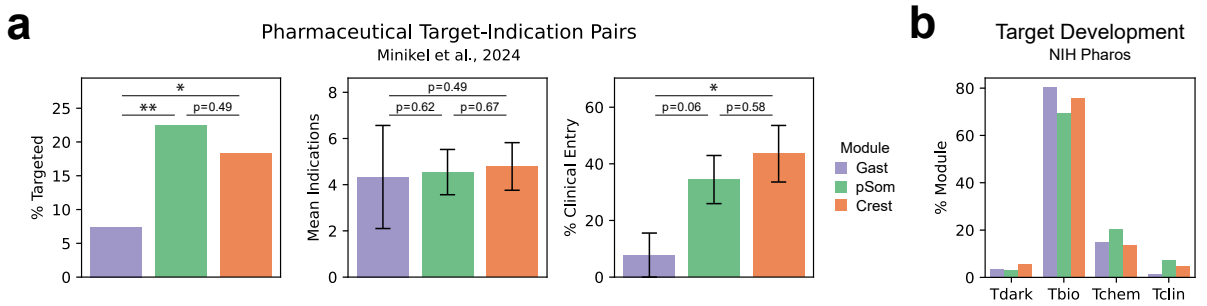
Supplementary Figure 4: Gene expression magnitude and timing correspond across e-EMT trajectories

(a, b) Scatter plots depict gene expression magnitude (a) and timing (b) for the 237 genes comprising MesenCoder's input features, compared pairwise across the gastrulation, posterior somitogenesis, and neural crest migration trajectories. Labels denote genes with the largest absolute residuals from the OLS fits (red lines), and correlations (red annotations) encompass all plotted genes.



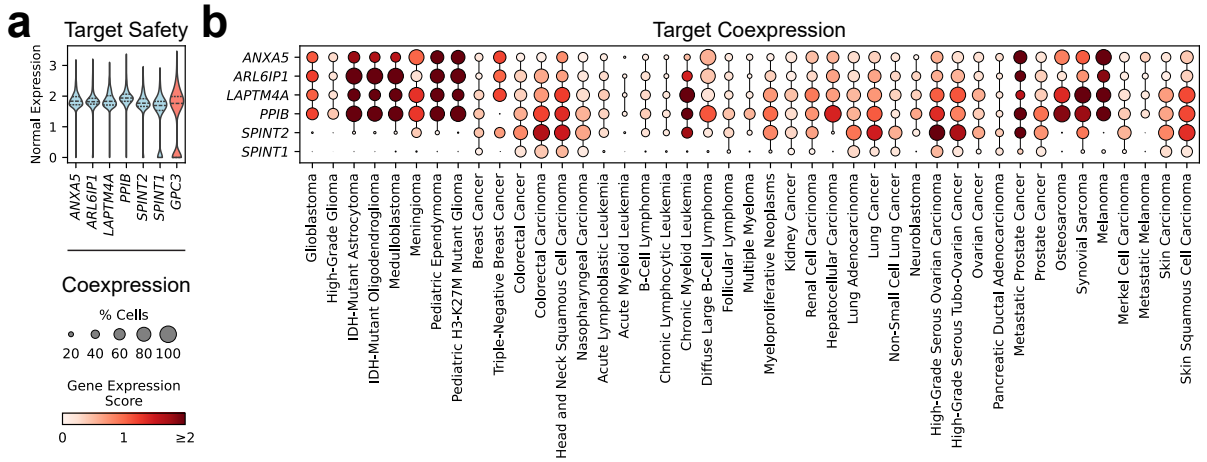
Supplementary Figure 5: HNSCC sample integration

UMAP embedding of malignant cells from head and neck squamous cell carcinoma (Puram *et al.*, 2017), integrated across samples and colored by sample.



Supplementary Figure 6: Drug development programs have neglected e-EMT modules

(a) Bar plots showing the proportion of each module's proteins with oncology target-indication pairs (left), their mean number of indications (middle), and their proportion entering clinical trials (right) (Minikel *et al.*, 2024). Error bars denote the standard error of the mean. *, $P < 0.05$; **, $P < 0.01$. **(b)** Grouped bar plot showing the proportion of each module's proteins assigned to Pharos target development levels (Kelleher *et al.*, 2023). Tdark, unstudied; Tbio, biologically characterized; Tchem, bound by potent small molecules; Tclin, targeted by an approved drug.



Supplementary Figure 7: Nominated targets' normal expression and pan-cancer coexpression

(a) Violin plot showing the expression of the nominated targets (light blue) and *GPC3* (light red) across normal cell types from postembryonic tissues (CZI Single-Cell Biology Program *et al.*, 2023). Nominated targets exhibit maximum normal expression below that of *GPC3*, a clinically tolerated vaccine target. Dashed lines indicate the median and upper/lower quartiles. **(b)** Dotplot showing the relative expression of nominated targets across 3CA diseases (Tyler *et al.*, 2025). Vertical lines denote target coexpression.